The Relative Efficiency of Stocks and Options under Ambiguity

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Abstract

This paper analyzes the relative cost of stock-based versus option-based incentive contracts in a principal-agent framework under model uncertainty. By embedding the monotone mean-variance ambiguity framework of Maccheroni et al. (2013) into the classical benchmark of Feltham and Wu (2001), we relax the traditional Bayesian paradigm to account for an agent who exhibits both risk and smooth ambiguity aversion. We demonstrate that the classical dominance of stocks over options is remarkably robust, persisting whenever model uncertainty is restricted strictly to volatility or when risk aversion dominates ambiguity aversion. However, we identify a novel, continuous characterization for cost reversal: option compensation becomes more efficient than stocks if and only if ambiguity simultaneously distorts both the mean and variance, and the agent is near-risk-neutral but highly ambiguity-averse. Under these conditions, the convex payoff structure of options filters out inter-model uncertainty by truncating low-mean distributions. Our results qualify the Maxmin findings of Liu et al. (2018), reconciling static incentive design with dynamic insights on robust contracting.

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1 Introduction

Modern organizations extensively utilize performance-based executive compensation schemes to mitigate the agency conflict and align managerial actions with shareholder wealth maximization. Within this framework, a firm's equity value serves as the primary informative signal for incentive design. As established by the seminal benchmark of Feltham and Wu (2001), performance-contingent mechanisms are broadly categorized into stock-based and option-based contracts. Assuming a classical mean-variance framework under objective probabilities, Feltham and Wu (2001) derive a foundational dominance result: whenever the agent has limited control over the firm's operating volatility, stock grants induce implementable effort at a strictly lower implementation cost to the principal than option-based structures.

However, the traditional Bayesian paradigm underlying this benchmark assumes that all economic uncertainty can be fully quantified via unique subjective or objective probability measures. As Gilboa and Marinacci (2016) and the empirical legacy of the Ellsberg paradox (Ellsberg, 1961) highlight, this assumption breaks down in environments characterized by structural model uncertainty, where the decision-maker cannot precisely identify the true data-generating process. Under model uncertainty, an agent exhibits ambiguity aversion—a distinct behavioral preference for known over unknown probability distributions. Because executive compensation is frequently deployed in highly ambiguous settings, such as early-stage technology sectors or rapidly shifting regulatory landscapes, the standard expected utility approximation may systematically misprice the latent risk and ambiguity premiums demanded by the agent.

To address these limitations, a prominent strand of the mechanism design literature has embraced non-Bayesian, worst-case objectives to analyze contract robustness. Within

this paradigm, Carroll (2015) provides a foundational benchmark by showing that when the principal faces nonquantifiable uncertainty regarding the agent's available action space, a strictly linear contract is uniquely optimal to guarantee robust performance. In the specific domain of corporate incentives, Liu et al. (2018) shifted the worst-case focus toward distribution-free environments by introducing ambiguity aversion into the stock-versus-options debate via the Maxmin Expected Utility (MEU) framework of Gilboa and Schmeidler (1989). They identified a striking reversal condition: under extreme ambiguity aversion where the agent evaluates contracts solely against worst-case volatility scenarios, the convexity of options can lower implementation costs relative to stocks by capitalizing on the agent's variance fears. Yet, while Carroll (2015) relies on technological ignorance to enforce linear structures, the binary nature of the MEU framework in Liu et al. (2018) restricts ambiguity exclusively to the variance parameter and forces the agent to place full probability mass on a single worst distribution, potentially overstating the contract's real-world distortion.

This paper relaxes these restrictive assumptions by embedding the smooth ambiguity framework of Klibanoff et al. (2005) and its monotone mean-variance extension by Maccheroni et al. (2013) into the contractual structure of Feltham and Wu (2001). This approach offers a natural parametric advantage: it separates risk aversion from ambiguity aversion and admits a continuous distribution of alternative models, moving beyond extreme, non-differentiable worst-case evaluations in both action and distribution spaces.

Our contribution to the mechanism design and incentives literature is threefold. First, we provide a generalized parametric characterization of executive compensation under smooth ambiguity, avoiding the binary restrictions of traditional Maxmin models. Second, we demonstrate that the classical dominance of stocks is remarkably robust and is structurally restored whenever model uncertainty is restricted solely to the volatility of the firm's terminal value. This static result directly reconciles incentive design with the dynamic insights of Hansen (2021) on robust contracting under time-varying volatility. Third, we establish a novel global sufficient threshold for cost reversal. We prove that options outperform stocks if and only if ambiguity distorts both the mean and the variance

simultaneously, and the agent is near-risk-neutral but highly ambiguity-averse. Under this precise interaction, the convex payoff profile of an option serves as an informational filter, truncating the agent's exposure to low-mean models.

By quantifying these trade-offs, we bridge classical incentive theory with the robust decision-making literature. We provide a tractable framework to evaluate compensation efficiency in corporate environments characterized by severe structural uncertainty, providing concrete support for the empirical calibrations observed in financial markets (Hara & Honda, 2022).

2 Incentive Design and the Agency Problem: Theory and Evidence

The design of optimal compensation structures lies at the confluence of behavioral motivations and structural risk allocation. Traditional microeconomic formulations often rest on the premise that linear or monotonic monetary transfers are sufficient to resolve moral hazard. However, a multi-disciplinary literature documents that high-powered incentives operate within complex psychological boundaries. For instance, behavioral interventions such as nonstandard framing can alter effort response curves where standard monetary rewards occasionally backfire (Kamenica, 2012). This friction is fundamentally theorized by Bénabou and Tirole (2003), who formalize the endogenous crowding-out of intrinsic motivation by extrinsic rewards. In their framework, performance-contingent pay may carry hidden psychological costs, signals negative informational attributes from the principal, and reduces long-term effort supply. Complementing this behavioral dimension, goal-setting paradigms emphasize that specific, exogenous objectives fundamentally alter an agent's task-performance function (Locke & Latham, 2002).

Despite these behavioral constraints, monetary incentives remain the primary operational mechanism for mitigating corporate agency frictions. When properly scaled, variable compensation yields substantial productivity dividends, as dropping incentive intensity too close to zero can induce severe behavioral shirking (Gneezy & Rustichini, 2000). In

the classical principal-agent paradigm established by Holmstrom and Milgrom (1994), the compensation contract is tasked with a dual optimization role: driving productivity while maintaining efficient risk-sharing. This trade-off becomes acute when a risk-averse manager must allocate attention across multi-task environments, forcing the firm to choose among diverse instruments including performance bonuses, debt-like benefits, or convex equity vehicles such as stocks and options.

Beyond motivating marginal effort, variable pay structures exert a powerful sorting effect on workforce composition. As proved by Lazear (1986), asymmetric pay schedules induce self-selection: adverse selection forces low-ability agents toward fixed salaries, whereas high-ability workers seek piece-rate schedules. This theoretical sorting is heavily validated by empirical parameters. Transitioning from fixed salaries to piece rates has been shown to trigger output expansions of 44% in industrial settings (Lazear, 2000), with at least half of the gains attributable to worker self-selection. Similar productivity expansions—ranging from 20% to 22.6%—are documented across field experiments and payroll records, despite minor trade-offs in output quality control (Paarsch & Shearer, 2000; Shearer, 2004).

From the employee's perspective, accepting a performance-contingent contract generates an endogenous wage premium. Empirical estimates isolate an average earnings expansion of 7% to 14% across manufacturing sectors (Lazear, 2000; Seiler, 1984), alongside structural wage increases of 6% for women and 9% for men in broader labor panels (Booth & Frank, 1999). Seiler (1984) demonstrates that this premium cannot be interpreted merely as a return to human capital; rather, it constitutes an equilibrium risk premium required to offset the heightened income variance that risk-averse workers must absorb.

When firms augment compensation contracts with explicit goal-setting mechanisms to anchor performance benchmarks, they attempt to minimize reward variance and signal strategic priorities. Effective goal-setting relies on distinct operational channels: challenging targets maximize effort margins, specific deadlines accelerate work velocity, and feedback loops sustain commitment (Latham & Locke, 2007; Locke & Latham, 2002). Nonetheless, structural misspecifications in target design generate severe corporate distortions. Setting

easy or ill-defined benchmarks invites strategic manipulation and minimal effort alignment, while unachievable thresholds trigger abrupt drops in managerial commitment (Latham & Yukl, 1975). Furthermore, in multi-period environments, the frequency and timing of evaluation windows dictate the momentum of effort, requiring tight feedback loops to manage information decay.

In sum, while the empirical consensus establishes the potency of variable pay and goal-setting in driving corporate productivity, the underlying theoretical architectures remain anchored to a restrictive informational baseline. This entire body of literature assumes that the precise probability distributions governing output variance are common knowledge to both the principal and the agent. Our framework breaks from this paradigm. We extend this foundational debate by asking how the strategic trade-off between stocks and options is transformed when economic agents confront structural model uncertainty, shifting the analysis from pure objective risk to the richer, less predictable domain of smooth ambiguity.

3 Theoretical Foundations: Benchmarks and Ambiguity Frameworks

This section establishes the analytical baseline required to evaluate the relative cost of incentive mechanisms under alternative informational paradigms. We begin by operationalizing the classical expected utility benchmark derived from the foundational framework of Feltham and Wu (2001), which validates the structural dominance of stock grants under objective probabilities. We then formalize the extreme ambiguity aversion benchmark via the Maxmin Expected Utility (MEU) framework of Liu et al. (2018), highlighting how worst-case decision-making can structurally reverse the classical result when uncertainty is confined to variance. Finally, we introduce the smooth ambiguity model of Klibanoff et al. (2005) and its monotone mean-variance formulation by Maccheroni et al. (2013). This progressive exposure serves a dual role: it maps the precise mathematical channels through which information architectures distort contract design and delineates how our

generalized approach encompasses existing paradigms as specialized limiting cases.

3.1 The Expected Utility Benchmark: Stocks versus Options

The baseline principal-agent architecture follows the foundational structure of Feltham and Wu (2001), wherein a risk-neutral principal contracts on behalf of firm equity-holders with a risk-averse manager over a single operational period. At $t = 0$, the principal offers a performance-contingent contract c. At $t = 1$, the firm's terminal gross value $\tilde{x}$ materializes as a normally distributed random variable. The objective probability distribution of $\tilde{x}$ depends on the manager's unobservable effort choice $a \in \mathbb{A}$, mapping an aggregate mean $m_x(a)$ and standard deviation $\sigma_x(a)$.

Although shareholders retain residual claims on the firm's net value, managerial incentives are typically generated through convex payoff structures consisting of a fixed wage combined with a grant of options characterized by an exercise price $k \geq 0$. Under this schedule, the contingent terminal transfer to the agent is proportional to $\max\{\tilde{x} - k, 0\}$. Pure stock compensation emerges as a specialized linear limiting case of this contract where the exercise price is set exactly to the boundary $k = 0$. Utilizing the properties of a truncated normal distribution (Greene, 2002), the conditional expectation $m_k(a)$ and intra-model variance $\sigma_k^2(a)$ of the option vehicle at $t = 1$ are characterized by:

$$m_k(a) = \sigma_x(a) \left[\phi(\xi) - \xi(1 - \Phi(\xi))\right] \tag{3.1}$$

$$\sigma_k^2(a) = \sigma_x^2(a) \left[(1 - \Phi(\xi) - \phi^2(\xi)) - \phi(\xi)(2\Phi(\xi) - 1)\xi + \Phi(\xi)(1 - \Phi(\xi))\xi^2\right] \tag{3.2}$$

where the underlying informational metrics and auxiliary variables are defined as:

$$\begin{aligned}\xi &= \frac{k - m_x(a)}{\sigma_x(a)},\\ \phi(\xi) &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\xi^2\right),\\ \Phi(\xi) &= \int_{-\infty}^{\xi} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}t^2\right) dt.\end{aligned}$$

The standardized metric ξ captures the precise distance, measured in standard deviations, between the contractually specified exercise price k and the expected firm value $m_x(a)$, while $\phi(\xi)$ and $\Phi(\xi)$ denote the standard normal probability density and cumulative distribution functions, respectively. As proved by Feltham and Wu (2001), both the expected value and the variance of an option contract scale monotonically with m_x and σ_x, but decay strictly as the exercise price k shifts rightward.

The manager selects an interior effort allocation $a \in \mathbb{A}$ to maximize a standard mean-variance certainty-equivalent (CE) function defined over his net wealth $\tilde{w} - g(a)$, where $g(a) = \frac{1}{2}a^2$ represents a strictly convex, quadratic disutility of effort:

$$CE(a, c) = E[\tilde{w} \mid a, c] - \frac{1}{2}\lambda \mathrm{Var}[\tilde{w} \mid a, c] - g(a) \tag{3.3}$$

where the parameters obey $\lambda > 0$, identifying the absolute risk-aversion coefficient.

The risk-neutral principal minimizes the total implementation cost required to induce a target effort level a, subject to the agent's voluntary participation and incentive compatibility constraints. This optimization problem is dual to net value maximization and is expressed as:

$$\begin{aligned}\underset{a,c}{\text{minimize}} \quad & E[\tilde{w} \mid a, c]\\ \text{subject to} \quad & CE(a, c) \geq CE^0\\ & a \in \underset{\hat{a} \in \mathbb{A}}{\operatorname{argmax}}\, CE(\hat{a}, c),\end{aligned} \tag{3.4}$$

where CE^0 denotes the agent's exogenous reservation utility. Under this specific baseline, Feltham and Wu (2001) derive the core benchmark result regarding the efficiency of stock-based structures:

Proposition 3.1. *Assume the manager is strictly risk-averse ($\lambda > 0$), effort shifts the mean of $\tilde{x}$ linearly but exerts no influence on its underlying volatility ($\frac{d\sigma_x}{da} = 0$), and the disutility of effort is quadratic. If the compensation schedule consists of a fixed wage combined with an option grant with exercise price $k \geq 0$, the implementation cost of inducing any target effort level a scales strictly and monotonically with k. Consequently, the total contract cost is globally minimized at the linear boundary $k = 0$, confirming the absolute dominance of stock-based incentives.*

3.2 Decision Theory and the Limits of the Bayesian Paradigm

The foundational contract architecture of Feltham and Wu (2001) rests entirely upon the subjective Bayesian paradigm. As conceptualized by Gilboa and Marinacci (2016), this paradigm assumes that economic agents reduce all structural uncertainty to single, unique subjective or objective probability measures. Under this restrictive informational baseline, a decision-maker chooses among alternative acts by evaluating expected utilities computed over a sharp prior distribution. However, this framework fails to characterize environments where the agent faces true Knightian uncertainty or epistemic ignorance regarding the parameters of the data-generating process. Forcing an agent to assign specific point probabilities under a complete lack of statistical consensus can generate purely arbitrary behavioral choices. Furthermore, the classical expected utility model relies fundamentally on the *independence axiom*, which dictates that a preference ordering over a space of acts remains invariant under convex combinations with a common third alternative. This axiomatic restriction is inherently inadequate for contractual environments with model uncertainty, where mixing independent profiles acts as an endogenous hedging mechanism that structurally collapses the volatility of expected returns.

The empirical and behavioral breakdown of the single-prior paradigm was famously established by the Ellsberg paradox (Ellsberg, 1961). Through experimental urn configurations, Ellsberg (1961) documented a systematic preference for known over unknown probability distributions, even when first and second objective moments were perfectly controlled. This behavioral friction revealed that economic agents do not merely price the

variance of payoffs (pure risk); they demand an additional equilibrium premium for their lack of confidence in the underlying probabilistic model, bringing the concept of *ambiguity aversion* to the core of decision theory.

To operationalize these behavioral insights within mechanism design, Gilboa and Schmeidler (1989) formalize the *Maxmin Expected Utility* (MEU) framework. This axiomatic approach replaces the single prior with a compact set of non-unique subjective priors, modeling a decision-maker who evaluates any act based on its minimum expected utility across all candidate distributions. By evaluating contracts exclusively against worst-case models, the MEU framework offers a tractable, albeit extreme, representation of worst-case caution.

Nevertheless, modeling an agent's behavioral response to model uncertainty as purely extreme oversimplifies the spectrum of professional executive behavior. To relax this binary evaluation, Klibanoff et al. (2005) introduced the *smooth ambiguity* framework (KMM), which admits a continuous subjective prior over the space of alternative probability measures. The operational superiority of the KMM structure stems from its parametric flexibility: it successfully untangles absolute risk aversion (governed by the curvature of the utility function) from smooth ambiguity aversion (governed by the curvature of a distinct evaluation function). Within this framework, the rigid Maxmin model of Gilboa and Schmeidler (1989) emerges merely as a limiting case where the ambiguity aversion coefficient approaches infinity. This smooth characterization provides a powerful mathematical toolkit to isolate how continuous changes in model uncertainty distort contract implementation.

In dynamic environments, this paradigm has been recently extended by Hansen (2021), who develops preference formulations that penalize unknown, time-varying volatility paths under repeated moral hazard. A vital insight from Hansen (2021) is that ambiguity regarding volatility alone—independent of uncertainty affecting the drift or mean parameter—yields highly specialized implications for the design of robust incentives. As established in Section 4, a parallel informational boundary operates within our static framework: when model uncertainty is restricted exclusively to the variance of the firm's terminal value, the classical stock-dominance result of Feltham and Wu (2001) is fully

restored under smooth preferences, clarifying the boundaries of contractual reversals.

3.3 Extreme Ambiguity Aversion: The Maxmin Approach

To formalize the benchmark for contractual reversals under extreme model uncertainty, Liu et al. (2018) embed a mean-variance Maxmin framework into the principal-agent architecture of Feltham and Wu (2001). In this setup, the principal targets an implementable effort level a by offering an equity-based contract $\tilde{w}$ to a risk- and ambiguity-averse manager whose localized preferences, conditional on a known volatility parameter, map to the certainty-equivalent formulation:

$$E[u] = E[\tilde{w}] - \frac{1}{2}\lambda \mathrm{Var}[\tilde{w}] - g(a) \tag{3.5}$$

where $g(a) = \frac{1}{2}a^2$ represents the quadratic disutility of effort, and λ denotes the absolute risk-aversion coefficient. The risk-neutral, ambiguity-neutral principal minimizes total executive compensation costs, which is dual to solving:

$$\begin{aligned} \underset{\tilde{w}}{\text{maximize}} \quad & E[\tilde{x} - \tilde{w}] \\ \text{subject to} \quad & E[u(a)] \geq 0 \\ & a \in \underset{\hat{a}\in\mathbb{A}}{\operatorname{argmax}}\, E[u(\hat{a})]. \end{aligned} \tag{3.6}$$

This baseline formulation maps directly to the standard benchmark problem (3.4) under a normalized reservation utility $CE^0 = 0$. The terminal gross value of the firm is parameterized as a linear production function perturbed by a homoscedastic shock, $\tilde{x} = a + l + \epsilon$, where $\epsilon \sim N(0, \sigma^2)$ and l represents an exogenous constant. Under these initial conditions, Liu et al. (2018) restate the classical stock-efficiency result:

Proposition 3.2. *Let C_s and C_k denote the total implementation costs required to induce a target effort level a via pure stock and option contracts, respectively. For any strictly positive option exercise price $k > 0$, the operational boundary satisfies $C_s < C_k$.*

The core structural departure introduced by Liu et al. (2018) involves imposing

uncertainty over the variance parameter σ^2. Utilizing the Maxmin Expected Utility (MEU) framework of Gilboa and Schmeidler (1989), they model an agent who acts under a compact set of non-unique variance priors bounded by an interval $\sigma \in [\sigma_1, \sigma_2]$. Evaluating the compensation schedule exclusively against the worst-case scenario, the agent's behavioral objective collapses to the following maxmin optimization problem:

$$\max_{a \in \mathbb{A}} \min_{\sigma \in [\sigma_1, \sigma_2]} E[u(a) \mid \sigma]. \tag{3.7}$$

The principal's risk perception is assumed to be captured by a sharp background probability measure $\sigma_p \in [\sigma_1, \sigma_2]$. In fact, because the principal can diversify structural risk across market portfolios, she remains ambiguity-neutral. This informational asymmetry between the principal and the agent alters the contract's implementation costs through a worst-case risk-perception shift:

Lemma 3.3. *Let σ_a and σ_p denote the subjective risk perceptions of the agent and the principal, respectively.*

(i) Under pure stock compensation ($k = 0$), the agent's worst-case variance evaluation binds at the upper bound $\sigma_a = \sigma_2$, yielding a total implementation cost of $C_s = \frac{1}{2}\lambda a^2 \sigma_2^2$.

(ii) Under an option-based schedule with exercise price $k > 0$ configured to induce effort a, if the agent's absolute risk aversion satisfies the upper-bound threshold

$$\lambda < \min_{\sigma \in [\sigma_1, \sigma_2]} \frac{2}{\beta_k(\sigma_1)} \frac{m_k(\sigma) - m_k(\sigma_1)}{\sigma_k^2(\sigma) - \sigma_k^2(\sigma_1)}, \tag{3.8}$$

then the agent's endogenous worst-case evaluation shifts abruptly to the lower variance bound σ_1. The resulting cost to the principal is parameterized as $C_k = \frac{1}{2}\lambda \beta_k^2(\sigma_1)\sigma_k^2(\sigma_1) + \beta_k(\sigma_1)[m_k(\sigma_p) - m_k(\sigma_1)]$.

The technical intuition underlying Lemma 3.3 stems from the structural asymmetry of payoff profiles. While a linear stock contract exposes the agent monotonically to variance extensions, the convex payoff structure of an option generates an option-gamma effect,

whereby the option's sensitivity to price changes increases with volatility. Because the option's downside is truncated at the strike price k, a higher underlying variance expands the conditional expected payoff m_k by flattening the favorable upper tail of the distribution without penalizing the lower tail. If the agent's risk aversion is sufficiently low relative to this convexity benefit, the marginal increase in expected returns compensates for the variance premium. Consequently, the agent's fear shifts toward the lower volatility bound, where the option's upside potential is minimized, enabling a potential reversal of the classical dominance result:

Proposition 3.4. *Suppose the lower model boundary σ_1 is sufficiently small to satisfy the initial dispersion condition $a^2\sigma_2^2 > \beta_k^2(\sigma_1)\sigma_k^2(\sigma_1)$. If the agent's absolute risk aversion satisfies the joint constraint*

$$\frac{2\beta_k(\sigma_1)[m_k(\sigma_p) - m_k(\sigma_2)]}{a^2\sigma_2^2 - \beta_k^2(\sigma_1)\sigma_k^2(\sigma_1)} < \lambda < \min_{\sigma\in[\sigma_1,\sigma_2]} \frac{2}{\beta_k(\sigma_1)} \frac{m_k(\sigma) - m_k(\sigma_1)}{\sigma_k^2(\sigma) - \sigma_k^2(\sigma_1)}, \tag{3.9}$$

then option compensation (with exercise price $k > 0$) induces the target effort level a at a strictly lower implementation cost to the principal than a stock-based grant ($C_k < C_s$).

Although Proposition 3.4 isolates a parametric region where options dominate stocks, this reversal is constrained by two restrictive assumptions. First, the result depends on a tight interval of low risk aversion (λ). Second, it is a structural artifact of the binary MEU paradigm, which forces the agent to place all probability mass on the worst distribution, thereby neglecting the rich spectrum of intermediate models and smooth belief adjustments. This boundary underscores the core objective of our paper. By applying the smooth ambiguity architecture of Klibanoff et al. (2005) and its monotone mean-variance formulation by Maccheroni et al. (2013), we achieve a generalized characterization of model uncertainty. Our approach relaxes the extreme worst-case evaluation and disentangles pure risk aversion (λ) from smooth ambiguity aversion (θ). As demonstrated in the following sections, this decomposition underscores the robustness of stock-based structures and reveals that cost reversal is driven not merely by volatility intervals but by an interaction with mean dispersion across subjective models.

3.4 The Smooth Ambiguity Framework and Monotone Mean-Variance Preferences

To overcome the binary limitations of worst-case Maxmin models and establish a continuous parametric environment, we adopt the smooth ambiguity framework conceptualized by Klibanoff et al. (2005) (hereafter KMM). Under the KMM paradigm, a decision-maker's preference ordering over a space of acts is represented by a double expectational functional form:

$$V(f) = \int_{\Delta} \phi \left(\int_{S} u(f) \, d\pi \right) d\mu, \tag{3.10}$$

where f represents a real-valued payoff function defined on a state space S, u denotes a standard von Neumann–Morgenstern utility function capturing attitudes toward pure risk, and π is a localized probability measure mapping the states in S. Model uncertainty is governed by the second-order subjective prior μ, which defines the decision-maker's beliefs over the compact set of all candidate probability measures Δ. The behavioral orientation toward structural ambiguity is isolated by the curvature of the evaluation function ϕ, where a strictly concave ϕ reflects ambiguity aversion. This formulation untangles risk from ambiguity within separate parameters, collapsing to the strict Maxmin expected utility framework of Gilboa and Schmeidler (1989) as a limiting case when the ambiguity aversion coefficient embedded in ϕ approaches infinity.

While the structural functional in (3.10) offers powerful foundational properties, its multi-layered expectational form can become mathematically intractable when embedded into asymmetric principal-agent optimizations. To operationalize this framework within corporate mechanism design, we leverage the local quadratic approximations derived by Maccheroni et al. (2013). Recall that for a classic utility maximizer evaluating wealth w perturbed by an investment shock h under a unique objective distribution P, the standard Arrow-Pratt second-order expansion yields the local certainty-equivalent:

$$c(w + h, P) \approx w + E_P[h] - \frac{1}{2}\lambda_u(w)\sigma_P^2(h), \tag{3.11}$$

where $\lambda_u(w) = -u''(w)/u'(w)$ denotes the Arrow-Pratt coefficient of absolute risk aversion,

rendering the variance-proportional risk premium that anchors traditional mean-variance objective functions:

$$U(f) = E_P[f] - \frac{1}{2}\lambda\sigma_P^2(f), \tag{3.12}$$

with $f = w + h$ and $\lambda = \lambda_u(w)$.

To extend this local tractability to environments with structural model uncertainty where the true data-generating process P is unknown, Maccheroni et al. (2013) model an agent who evaluates a family of alternative probability distributions $Q \in \Delta$ via a secondary evaluation utility v, representing attitudes toward model specification. The smooth ambiguity certainty-equivalent under this configuration maps to:

$$c(w + h) = v^{-1}\left(E_\mu\left[v\left(u^{-1}\left(E_Q[u(w + h)]\right)\right)\right]\right). \tag{3.13}$$

Whenever the curvature of v departs from the curvature of u, model uncertainty cannot be reduced to simple risk. By executing a joint second-order Taylor expansion on (3.13) around the institutional wealth baseline, Maccheroni et al. (2013) derive a sharp mean-variance-ambiguity certainty-equivalent:

$$c(w + h) \approx w + E_{\bar{Q}}[h] - \frac{1}{2}\lambda_u(w)\sigma_{\bar{Q}}^2(h) - \frac{1}{2}(\lambda_v(w) - \lambda_u(w))\sigma_\mu^2(E[h]), \tag{3.14}$$

where $\bar{Q} = \int_\Delta Q\, d\mu(Q)$ defines the reduced, predictive probability measure across the model space, and $E[h] : \Delta \to \mathbb{R}$ behaves as a random variable mapping each subjective candidate model Q to its conditional expected value $E_Q[h]$. The underlying statistical metric $\sigma_\mu^2(E[h])$ isolates the inter-model variance of expected payoffs across the alternative distributions. Scaled by the divergence between the dual aversion coefficients $(\lambda_v - \lambda_u)$, this term formalizes the localized *ambiguity premium*.

By mapping the asset profiles to $f = w+h$, defining the absolute risk aversion parameter as $\lambda = \lambda_u(w)$, establishing the absolute ambiguity aversion metric as $\theta = \lambda_v(w) - \lambda_u(w)$, and setting the reference predictive measure to $P = \bar{Q}$, we obtain the augmented, monotone

mean-variance preference function that governs our principal-agent framework:

$$U(f) = E_P[f] - \frac{1}{2}\lambda\sigma_P^2(f) - \frac{1}{2}\theta\sigma_\mu^2(E[f]). \tag{3.15}$$

Under this generalized preference architecture, the parameters λ and θ separately isolate the structural weights assigned to intra-model risk and inter-model ambiguity, respectively.

The economic validity and empirical identifiability of this linear-quadratic calibration are strongly corroborated by Hara and Honda (2022). In their asset pricing formulations, they demonstrate that under smooth model uncertainty, ambiguity-averse market participants deviate systematically from traditional mean-variance efficient frontiers. By construction, the ambiguity parameter θ formalized in (3.15) can be backed out and quantified using empirical equity returns. This empirical validation ensures that our theoretical contract comparisons between stock grants and option vehicles are not merely abstract exercises, but rest upon an analytically quantifiable and behaviorally sound foundation.

4 Contract Design under Monotone Mean-Variance Ambiguity

This section develops our core principal-agent optimization model by embedding the monotone mean-variance framework of Maccheroni et al. (2013) into the asymmetric incentive structure of Feltham and Wu (2001). While classical agency formulations restrict the information architecture to a single, sharp objective probability distribution, our framework explicitly formalizes the agent's epistemic uncertainty regarding the underlying stochastic process that governs the firm's terminal value. This generalization bypasses the binary, non-differentiable worst-case logic inherent to strict Maxmin expected utility models, providing a continuous parametric characterization of how structural model uncertainty distorts contract execution and effort implementation.

4.1 Implementation Costs and Augmented Certainty Equivalents

The operational point of departure for this economic analysis requires deriving the agent's augmented certainty equivalent under non-unique priors. Under the monotone mean-variance mapping established by Maccheroni et al. (2013), the manager's baseline certainty equivalent is penalized by two distinct structural components: intra-model variance, which captures traditional economic risk, and inter-model dispersion of expected conditional returns, which isolates smooth ambiguity. This dual penalty architecture provides a tractable foundation to evaluate the implementation costs of alternative stock- and option-based compensation schedules under an expanded support of beliefs, directly capturing the agent's psychological and behavioral discount for imprecise probabilistic information.

Let $\tilde{x}$ denote the random gross terminal value of the firm realized at $t = 1$. Structural ambiguity is operationalized via a finite family of alternative normal distributions, where each model scenario $j \in \{1, 2, \ldots, n\}$ is assigned a subjective second-order prior probability μ_j. This informational hierarchy is formalized in Figure 1.

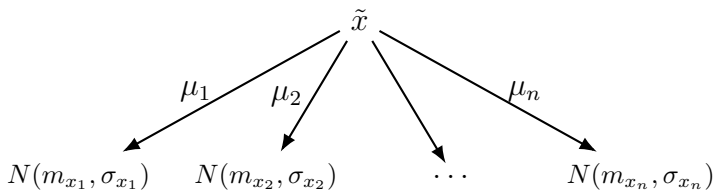


Figure 1: Informational hierarchy: Ambiguity over the probability distribution of firm terminal value $\tilde{x}$.

Conditional on each localized scenario j, the firm value $\tilde{x}$ is normally distributed with an endogenous conditional mean $m_{xj}(a)$ and a conditional standard deviation $\sigma_{xj}(a)$. The manager's behavioral aversion operates on two distinct margins: the horizontal width of any single probability density curve reflects pure intra-model risk, whereas the structural displacement among the alternative candidate curves reflects the model uncertainty or inter-model ambiguity.

To operationalize the interaction between these dual margins, consider an explicit parameterized configuration characterized by three potential scenarios ($n = 3$). This parametric specification captures simultaneous model uncertainty over both the first and

second moments of the firm's terminal value, formalizing a pessimistic (μ_1), a neutral (μ_2), and an optimistic model (μ_3), as graphically illustrated in Figure 2.

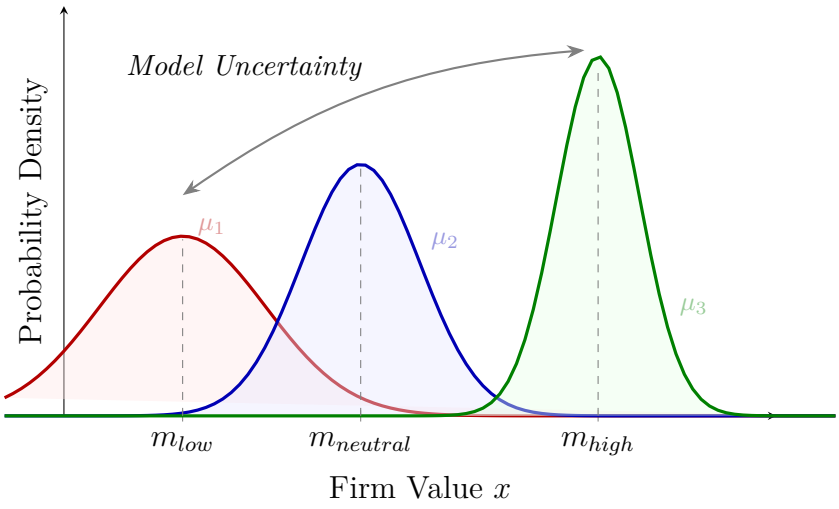


Figure 2: Parametric specification with three structural scenarios ($n = 3$). The horizontal variance of each independent profile represents **risk**, whereas the spatial displacement between the alternative density functions captures **ambiguity**.

Within this mathematical environment, the agent's augmented mean-variance certainty-equivalent function, conditional on a choice of effort a and a compensation structure c, is defined by:

$$CE(a, c) = E_P\left[\widetilde{w} \mid a, c\right] - \frac{1}{2}\lambda \mathrm{Var}_P\left[\widetilde{w} \mid a, c\right] - \frac{1}{2}\theta \mathrm{Var}_\mu\left(E_Q\left[\widetilde{w} \mid a, c\right]\right) - g(a) \tag{4.1}$$

where $P = E_\mu[Q] = \int_\Delta Q\, d\mu(Q)$ represents the predictive, reduced probability distribution across the entire model space, $g(a) = \frac{1}{2}a^2$ is the quadratic disutility of effort, and the structural coefficients $\lambda > 0$ and $\theta \geq 0$ parameterize absolute risk aversion and absolute smooth ambiguity aversion, respectively.

The risk-neutral and ambiguity-neutral principal maximizes the expected net terminal value of the firm, yielding the operational objective function:

$$\max_{a,c} E\left[\widetilde{x} - \widetilde{w} \mid a, c\right] \tag{4.2}$$

subject to the agent's voluntary participation and incentive compatibility constraints:

$$CE(a, c) \geq CE^0 \tag{4.3}$$

$$a \in \arg\max_{\widehat{a} \in \mathbb{A}} CE(\widehat{a}, c) \tag{4.4}$$

We parameterize the contract space via a compensation schedule $C_k(\alpha_k, \beta_k)$, wherein the manager's terminal payoff is characterized by the convex schedule $\tilde{w} = \alpha_k + \beta_k \max\{\widetilde{x} - k, 0\}$. Here, α_k defines the deterministic fixed base salary and β_k isolates the intensity of the option grant configured at a strike price $k \geq 0$.

Proposition 4.1. *Under the augmented monotone mean-variance framework, the agent's conditional certainty equivalent maps to:*

$$CE(a, c) = \alpha_k + \beta_k \overline{m}_k(a) - \frac{1}{2}\lambda \beta_k^2 \overline{\sigma}_k^2(a) - \frac{1}{2}\theta \, Var_\mu\Big(\alpha_k + \beta_k m_k(a)\Big) - g(a) \tag{4.5}$$

where the conditional mean $\overline{m}_k(a)$ and intra-model variance $\overline{\sigma}_k^2(a)$ evaluated relative to the predictive mixture distribution $\bar{P}$ are characterized by:

$$\overline{m}_k(a) = \overline{\sigma}_x(a) \left[\phi(\xi) - \xi(1 - \Phi(\xi))\right] \tag{4.6}$$

$$\overline{\sigma}_k^2(a) = \overline{\sigma}_x^2(a) \Big[(1 - \Phi(\xi) - \phi^2(\xi)) - \phi(\xi)(2\Phi(\xi) - 1)\xi + \Phi(\xi)(1 - \Phi(\xi))\xi^2\Big] \tag{4.7}$$

conditional on the aggregate state parameters $\overline{m}_x(a)$ and $\overline{\sigma}_x(a)$, with the standardized threshold ξ and normal distributions ϕ, Φ operating as defined in the baseline benchmark.

Because the contract parameters α_k and β_k are deterministic at $t = 0$, the second-order dispersion operator $\mathrm{Var}_\mu(\alpha_k + \beta_k m_k(a))$ simplifies strictly to $\beta_k^2 \sigma_\mu^2(m_k(a))$ via the translation invariance properties of variance. Consequently, the localized certainty equivalent collapses to the tractable parametric formulation:

$$CE(a, c) = \alpha_k + \beta_k \overline{m}_k(a) - \frac{\beta_k^2}{2}\Big(\lambda \overline{\sigma}_k^2(a) + \theta \sigma_\mu^2(m_k(a))\Big) - g(a) \tag{4.8}$$

Normalizing the manager's exogenous reservation utility to $CE^0 = 0$, optimization

forces the principal to bind the participation constraint (4.3) at the baseline boundary, isolating the equilibrium fixed wage structure:

$$\alpha_k = \frac{\beta_k^2}{2}\left(\lambda\overline{\sigma}_k^2(a) + \theta\sigma_\mu^2(m_k(a))\right) + g(a) - \beta_k\overline{m}_k(a) \tag{4.9}$$

where the inter-model uncertainty mapping the variance of the expected option payoffs across alternative distribution models is formalized as:

$$\sigma_\mu^2(m_k(a)) = \sum_{j=1}^{n} \mu_j \left(m_{kj}(a) - \overline{m}_k(a)\right)^2 \tag{4.10}$$

The structural fixed wage in (4.9) serves as an equilibrium transfer, compensating the agent for his localized reservation utility, the intra-model risk premium, the smooth inter-model ambiguity premium, and the convex disutility of effort, net of the expected baseline value of the derivative vehicles.[1]

Substituting the binding fixed wage schedule into the principal's objective function establishes the full implementation problem under asymmetric information.

Proposition 4.2. *The principal's programmatic optimization problem is characterized by:*

$$\max_{a,\beta_k,k}\left\{\overline{m}_x(a) - \left[\frac{\beta_k^2}{2}\left(\lambda\overline{\sigma}_k^2(a) + \theta\sigma_\mu^2(m_k(a))\right) + g(a)\right]\right\} \tag{4.11}$$

subject to the agent's incentive compatibility constraint:

$$a \in \arg\max_{\widehat{a}\in\mathbb{A}}\left\{\beta_k\overline{m}_k(\widehat{a}) - \frac{\beta_k^2}{2}\left(\lambda\overline{\sigma}_k^2(\widehat{a}) + \theta\sigma_\mu^2(m_k(\widehat{a}))\right) - g(\widehat{a})\right\} \tag{4.12}$$

Assuming a differentiable, interior solution space, the first-order condition (FOC) governing the agent's optimal effort choice a satisfies:

$$\beta_k\frac{d\overline{m}_k(a)}{da} - \frac{\beta_k^2}{2}\left(\lambda\frac{d\overline{\sigma}_k^2(a)}{da} + \theta\frac{d\sigma_\mu^2(m_k(a))}{da}\right) - g'(a) = 0 \tag{4.13}$$

[1] If the agent exhibits ambiguity neutrality ($\theta = 0$), the preference functional collapses to the classical expected utility model where the terminal value behaves as $\tilde{x} \sim N(\overline{m}_x, \overline{\sigma}_x)$. Under this limiting condition, $\overline{m}_x = \sum \mu_j m_{xj}$ represents the first moment, and the variance maps directly to the reduced mixture specification.

4.2 Optimal Effort and the Implementability Threshold

To evaluate whether option-based structures entail higher implementation costs than stock-based contracts, we solve for the equilibrium contract intensity, $\beta_k(a)$, required to implement a target effort level a. This optimization requires establishing the boundary conditions under which an arbitrary action a is structurally *implementable* within the augmented mean-variance-ambiguity framework.

Proposition 4.3. *Assume the manager's action space is one-dimensional, $a \in [0, \infty)$, and let effort induce a linear shift on the conditional mean of each candidate distribution $j \in \{1, \dots, n\}$, such that $m_{xj}(a) = m_j + a$. It follows by construction that the aggregate predictive mean satisfies $\overline{m}_x(a) = \overline{m} + a$, where $\overline{m} = \sum_{j=1}^{n} \mu_j m_j$. Under this informational structure, the conditional derivatives with respect to effort obey:*

$$\frac{d\overline{m}_k(a)}{da} = 1 - \Phi(\xi),$$
$$g'(a) = a.$$

A target action a is implementable if and only if the marginal disutility of effort does not exceed an endogenous global threshold $G(\xi)$ governed by the agent's dual aversion coefficients:

$$a \leq G(\xi) = \frac{(1 - \Phi(\xi))^2}{2\left(\lambda \frac{d\overline{\sigma}_k^2(a)}{da} + \theta \frac{d\sigma_\mu^2(m_k(a))}{da}\right)} \tag{4.14}$$

Whenever the implementability condition in (4.14) holds, the optimal contract intensity $\beta_k(a)$ is characterized by the unique minimum positive root of the agent's first-order condition (4.13), mapping strictly to:

$$\beta_k(a) = \frac{2a}{(1 - \Phi) + \sqrt{(1 - \Phi)^2 - 2a\left(\lambda \frac{d\overline{\sigma}_k^2(a)}{da} + \theta \frac{d\sigma_\mu^2(m_k(a))}{da}\right)}} \tag{4.15}$$

The parametric architecture of (4.15) demonstrates that the equilibrium option grant size $\beta_k(a)$ scales monotonically with the strike price k, the absolute risk-aversion parameter λ, and the smooth ambiguity-aversion coefficient θ. Conversely, the implementability

frontier $G(\xi)$ defined in (4.14) contracts strictly under increases in these same structural parameters.

The economic mechanism driving this contraction stems from the compounding of risk and model uncertainty. As the strike price k shifts rightward or the ambiguity coefficient θ increases, the manager absorbs not only heightened intra-model variance but also severe second-order volatility regarding the true data-generating process of the firm. For an agent exhibiting both risk and ambiguity aversion, expanding the volume of convex derivative vehicles does not yield indefinite effort expansions. At the margin, the geometric growth of the risk and ambiguity premiums eventually offsets the marginal utility derived from expected wealth. The introduction of model uncertainty ($\theta > 0$) severely tightens this incentive constraint; the marginal change in inter-model variance across the alternative probability distributions ($\frac{d\sigma_\mu^2}{da}$) acts as an additional behavioral penalty, restricting the set of implementable high-effort actions relative to the classical risk-only benchmark.

4.3 Efficiency Analysis: Stocks versus Options under Model Uncertainty

Pure stock compensation constitutes a linear limiting case of the general option-based schedule where the contractually specified strike price satisfies $k = 0$. Because the manager's target effort level a shifts the conditional mean of the gross terminal value without mutating the underlying volatility structure, the stock-contingent contract dynamics reduce strictly to $\overline{m}_0(a) = \overline{m} + a$, $\frac{d\overline{m}_0(a)}{da} = 1$, $\overline{\sigma}_0^2(a) = \overline{\sigma}^2$, and $\frac{d\overline{\sigma}_0^2(a)}{da} = 0$. By extension, the inter-model uncertainty mapping the stock payoff remains invariant to the choice of action, yielding the boundary condition $\frac{d\sigma_\mu^2(m_0(a))}{da} = 0$.

Imposing these parametric restrictions on the equilibrium incentive condition in (4.15) isolates the optimal volume of stocks required to implement the target effort level a:

$$\beta_0(a) = a \tag{4.16}$$

The identity in (4.16) demonstrates that the equilibrium stock grant size $\beta_0(a)$ is struc-

turally orthogonal to the firm's localized variance, absolute risk aversion, and smooth ambiguity aversion. Under this linear incentive architecture, the agent expands productive effort until the marginal utility of equity ownership equates to the marginal disutility of action, removing any endogenous barrier to contract implementability.

Conversely, for derivative contracts structured exactly at-the-money relative to the conditional predictive distribution ($\widehat{k} = \overline{m}_x(a)$), the standardized metric satisfies $\xi = 0$. This boundary constraint implies that the normal density functions collapse to the constants $\phi(0) = 1/\sqrt{2\pi}$ and $\Phi(0) = 1/2$. Substituting these specific functional values into the equilibrium incentive condition (4.15) characterizes the optimal option allocation:

$$\beta_k(a) = 2a \left[\frac{1}{2} + \sqrt{\frac{1}{4} - 2a \left(\lambda \frac{d\overline{\sigma}_k^2(a)}{da} + \theta \frac{d\sigma_\mu^2(m_k(a))}{da} \right)} \right]^{-1} \tag{4.17}$$

Under the specialized benchmark of simultaneous risk and ambiguity neutrality ($\lambda = \theta = 0$), the optimal grant intensity simplifies to:

$$\beta_k(a) = 2a \left[\frac{1}{2} + \sqrt{\frac{1}{4}} \right]^{-1} = 2a \tag{4.18}$$

The structural parameter in (4.18) establishes that even in the absence of behavioral aversions, the principal must issue exactly twice the volume of options relative to a pure stock grant to implement an identical effort level. This scaling property is an equilibrium requirement to offset the asymmetric truncation of the option payoff space, which collapses to zero whenever the realized terminal value falls short of the exercise price k. Whenever the manager exhibits risk or ambiguity aversion, the principal must expand $\beta_k(a)$ further to provide a compensatory premium for the heightened volatility of the derivative vehicle.

Comparing the total implementation costs between these alternative instruments reveals a fundamental economic trade-off. Although convex options generate higher high-powered incentives per unit, they amplify the agent's structural exposure to both intra-model risk and inter-model uncertainty. To analytically isolate this mechanism, we decompose the agent's aggregate compensation variance into two distinct structural metrics: the *intra-model risk* exposure (f_k^2), mapping volatility within each sub-distribution, and

the *inter-model uncertainty* exposure (y_k^2), mapping the dispersion across the candidate probabilistic measures.

Intra-model Risk: Defined as $f_k^2(a) = \beta_k^2(a)\overline{\sigma}_k^2(a)$. For an at-the-money schedule ($k = \widehat{k}$), the localized variance exposure satisfies the structural inequality:

$$f_k^2(a) = a^2\overline{\sigma}^2\left[4\left(\frac{1}{2} - \frac{1}{2\pi}\right)\right] > a^2\overline{\sigma}^2 = f_0^2(a) \tag{4.19}$$

The inequality in (4.19) confirms that the expanded grant volume required under options ($\beta_k > \beta_0$) causes option contracts to impose a strictly higher intra-model risk premium than linear stock grants.

Inter-model Uncertainty: Defined as $y_k^2(a) = \beta_k^2(a)\sigma_\mu^2(m_k(a))$. This metric formalizes how the agent's expected utility varies across the n alternative distributions. Utilizing the subjective second-order probability measure μ, we map the relative exposures under options and stocks:

$$y_k^2(a) = 4a^2\left(\sum_{j=1}^{n}\mu_j\left(m_{\widehat{k}j}(a) - \overline{m}_{\widehat{k}}(a)\right)^2\right) \tag{4.20}$$

$$y_0^2(a) = a^2\left(\sum_{j=1}^{n}\mu_j\left(m_{0j}(a) - \overline{m}_0(a)\right)^2\right) \tag{4.21}$$

The manager's aggregate exposure to smooth model uncertainty is strictly higher under an option-based contract whenever the parametric condition satisfies $y_k^2(a) > y_0^2(a)$. This structural condition binds whenever:

$$4\left(\sum_{j=1}^{n}\mu_j\left(m_{\widehat{k}j}(a) - \overline{m}_{\widehat{k}}(a)\right)^2\right) > \sum_{j=1}^{n}\mu_j\left(m_{0j}(a) - \overline{m}_0(a)\right)^2 \tag{4.22}$$

Because the expansion coefficient satisfies $4 > 1$ and the convex options capitalize on mean shifts across alternative distributions, the condition in (4.22) is widely satisfied in non-degenerate environments. This result confirms that option structures exacerbate implementation costs not merely via intra-model risk premiums but also through their hypersensitivity to model uncertainty. Grounded in this variance decomposition, we

establish a global sufficient condition for the structural dominance of stock-based incentive contracts:

Theorem 4.4. *Stock-based incentive mechanisms incur a strictly lower implementation cost to the principal than option-based mechanisms if the inter-model dispersion of conditional expected payoffs satisfies the global boundary condition:*

$$\sum_{j=1}^{n} \mu_j \left(m_{\widehat{kj}}(a) - \overline{m}_{\widehat{k}}(a) \right)^2 > \frac{1}{4} \sum_{j=1}^{n} \mu_j \left(m_{0j}(a) - \overline{m}_0(a) \right)^2 \tag{4.23}$$

The localized sensitivity of the inter-model uncertainty premium y_k with respect to marginal changes in the strike price k is structurally indeterminate. This indeterminacy is driven by a complex interaction between the option-grant derivative, $d\beta_k(a)/dk$, and the underlying covariance structure mapping expected payoffs across alternative probability measures. This structural boundary highlights the strategic importance of Theorem 4.4: rather than relying on localized, coordinate-dependent derivative tests, we establish a global sufficient threshold defined over the inter-model variance that guarantees the cost-efficiency and structural dominance of stock-based corporate incentives.

In the presence of model uncertainty ($n \geq 2$), the global sufficient condition in (4.23) is satisfied whenever the inter-model variance of expected payoffs generated by an option contract is at least one-quarter of the inter-model variance tracking a pure stock grant. Under a continuous scaling of alternative models ($n \to \infty$), the subjective probability weights μ_j smooth into a continuous density function, wherein the structural dominance of stocks remains bounded by this lower-bound dispersion ratio. The coefficient of 4 in the denominator represents the baseline geometric boundary derived from the specialized case of simultaneous risk and ambiguity neutrality. This factor scales strictly upward whenever the manager exhibits heightened behavioral aversions across either dimensional margin. Because the linear contract intensity $\beta_0(a)$ remains invariant to variations in λ and θ, whereas the convex option intensity $\beta_k(a)$ is strictly increasing in both coefficients, a higher aversion further contracts the principal's implementability region for option structures. For a risk- and ambiguity-averse manager, the cumulative compensation premium expands at

a faster rate under option-based schedules, reinforcing the global optimality of stock-based incentives.

Even if the structural dispersion condition in (4.23) is locally violated, the classical benchmark result remains robust provided the parametric spaces satisfy the joint inequality:

$$\lambda f_k^2(a) > \theta y_k^2(a), \quad \forall k \geq 0. \tag{4.24}$$

The condition in (4.24) dictates that in specialized environments where stock grants exhibit a higher inter-model dispersion profile ($y_k^2 < y_0^2$), they remain strictly less costly to implement if the intra-model risk premium (λf_k^2) dominates the corresponding smooth ambiguity premium (θy_k^2). Substituting the functional properties derived for at-the-money option vehicles transforms this behavioral relationship into the sharp inequality:

$$\lambda \left[\overline{\sigma}^2 \left(\frac{1}{2} - \frac{1}{2\pi} \right) \right] > \theta \left[\sum_{j=1}^{n} \mu_j (m_{kj}(a) - \overline{m}_k(a))^2 \right] \tag{4.25}$$

The formal structure of (4.25) yields two core economic insights. First, the relative efficiency of stock-based structures over option vehicles is strictly increasing in the behavioral ratio mapping absolute risk aversion to smooth ambiguity aversion (λ/θ). Second, stock grants are strictly dominant whenever structural model uncertainty is restricted exclusively to the variance parameter of the firm's terminal value. In the absence of uncertainty affecting the first moment of the distribution—such that $m_{kj}(a) = \overline{m}_k(a)$ for all candidate models j—the right-hand side of (4.25) collapses to zero by construction, guaranteeing that the dominance inequality holds for any non-degenerate risk-aversion coefficient $\lambda > 0$.

These boundaries delineate the precise parametric edge-case configuration under which the classical efficiency result reverses, causing stock-based contracts to become costlier to implement than option schedules. This cost reversal requires the simultaneous violation of the global bounds in (4.23) and (4.25), which is achievable if and only if the contracting environment satisfies three simultaneous constraints:

1. **Structural Model Asymmetry:** The inter-model variance over expected returns must be substantially higher under linear stock contracts than under option

structures.

2. **Coarse Model Specification:** The ambiguity must be restricted to a highly constrained set of potential distribution profiles, preventing the smooth convergence of the dispersion threshold.

3. **Dominant Ambiguity Aversion:** The manager must operate near risk-neutrality ($\lambda \to 0$) while exhibiting strict smooth ambiguity aversion ($\theta > 0$), isolating an asymmetric preference topology that eliminates the intra-model risk premium that structurally penalizes convex option contracts.

4.4 Numerical Illustration and Calibration

Evaluating the economic plausibility of the contract cost reversal requires executing a numerical calibration grounded in parameters sourced from the empirical asset pricing literature. Following the institutional parameters identified by Hara and Honda (2022), who calibrate the monotone mean-variance preference functional to U.S. equity data, we set the smooth absolute ambiguity aversion coefficient to $\theta = 45$. For the absolute risk-aversion margin, we adopt the conservative baseline parameter used in contemporary executive compensation models, setting $\lambda = 1$ (Liu et al., 2018).

This highly asymmetric parameter configuration—characterized by near-risk neutrality ($\lambda = 1$) coupled with high smooth ambiguity sensitivity ($\theta = 45$)—reflects a precise corporate reality common in high-growth, technology-driven sectors. While traditional agency models assume severe risk aversion due to managerial under-diversification, executives in early-stage ventures often behave as quasi-owners who internalize the firm's growth objectives, rendering them relatively insensitive to objective intra-model volatility. However, their behavioral sensitivity to structural model uncertainty (θ) remains highly pronounced. This aversion does not stem from personal caution, but rather captures severe informational limitations, poor baseline data, and the unique economic conditions under which the firm operates (e.g., unpredictable patent approval or regulatory timelines). Under these specific institutional constraints, the manager embraces objective risk but

demands a substantial equilibrium premium for the lack of confidence in the underlying probability model.

The simulation framework operates under a discrete scenario architecture composed of three potential models ($n = 3$). In high-uncertainty industrial environments—such as early-stage biotechnology ventures navigating unmapped clinical trial pathways—firm terminal equity is highly sensitive to binary, discontinuous R&D outcomes, generating high spatial dispersion across conditional expected means. Under this asymmetric belief structure, at-the-money option schedules contract the inter-model uncertainty premium (y_k^2) by structurally isolating the agent from low-mean distribution trajectories. Conversely, by construction, these same convex vehicles expand the intra-model risk premium (f_k^2) due to the mathematical properties of their asymmetric payoff profiles.

The calibrated cost breakdown is quantified and decomposed in Figure 3, utilizing a computational parameter scale where the baseline effort disutility maps to $g(a) = 10$ units. Under a parametric state space where the inter-model variance tracking linear stock grants is approximately 4.5 times greater than the variance under option alternatives, and the preference configuration reflects high relative ambiguity aversion ($\theta/\lambda = 45$), the total implementation cost borne by the principal under an option contract ($C_k = 98$) is 10.9% lower than the equilibrium cost of a pure stock grant ($C_s = 110$). This parametric configuration establishes that in corporate entities navigating severe structural model uncertainty, the classical dominance threshold of stock-based incentives breaks down.

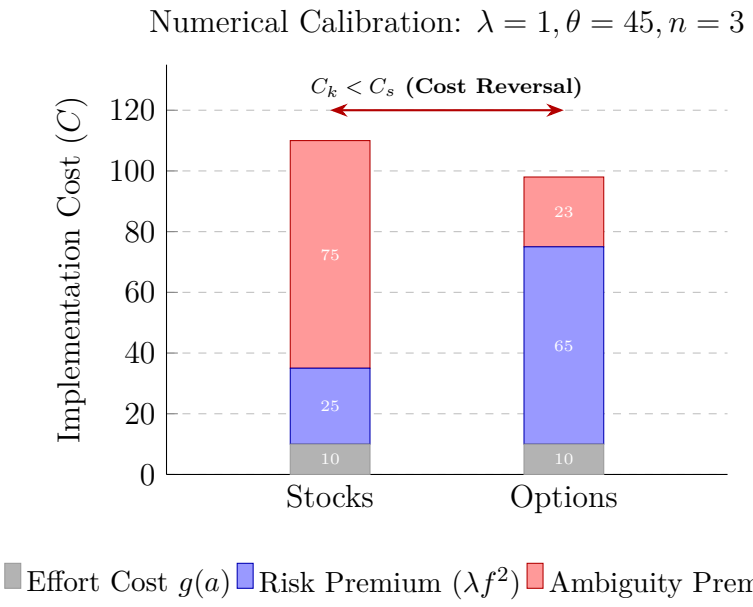


Figure 3: Equilibrium Cost Decomposition under Severe Model Uncertainty. While convex options impose a higher intra-model risk premium, the disproportionate inter-model ambiguity premium generated by linear stock schedules under high mean dispersion (σ_μ^2) triggers a structural cost reversal.

This numerical calibration delineates the foundational transmission mechanism that separates our findings from the Maxmin conclusions of Liu et al. (2018). While their MEU approach suggests that model uncertainty over volatility parameters is sufficient to justify option-based contract adoption, our monotone mean-variance framework proves that the option advantage is fundamentally driven by the spatial dispersion of conditional means across alternative probability measures (y_k^2 versus y_0^2). Under smooth ambiguity preferences, the convex payoff schedule of an option operates as an informational filter, compressing the manager's exposure to structural uncertainty by truncating the lower tail of the distribution. This isolation reveals a more nuanced behavioral channel for contract optimization: options achieve cost efficiency not as a pure consequence of macroeconomic volatility, but specifically when their capacity to mitigate second-order inter-model uncertainty expands faster than, and ultimately dominates, their inherently higher intra-model risk premium.

5 Conclusion

This paper has established a parametric analysis of the relative implementation costs of stock-based versus option-based incentive contracts under structural model uncertainty. By embedding the monotone mean-variance framework of Maccheroni et al. (2013) into the principal-agent architecture of Feltham and Wu (2001), we relax the classical single-prior Bayesian restriction to account for an agent who exhibits both risk aversion and smooth ambiguity aversion. The central finding demonstrates that the classical benchmark result—wherein linear stock grants dominate convex option structures by inducing effort at a lower cost to the principal—is robust, persisting across the vast majority of the parametric space. Concurrently, we isolate a specific configuration under which this dominance paradigm breaks down, providing a continuous characterization of contractual reversals.

The strategic trade-off between stocks and options under smooth ambiguity is governed by the relative weights assigned to two distinct variance metrics: intra-model risk (f_k^2), capturing volatility within independent distributions, and inter-model uncertainty (y_k^2), capturing the spatial dispersion across alternative probability measures. As characterized in the efficiency mapping in Figure 4, stock dominance is structurally preserved throughout Region I. This baseline efficiency is maintained whenever the inter-model variance satisfies the global sufficient condition in Theorem 4.4, or whenever the absolute risk-aversion coefficient dominates the smooth ambiguity margin ($\lambda f_k^2 > \theta y_k^2$). Under the parameterization where model uncertainty is restricted exclusively to the volatility parameter—as formulated in the Maxmin expected utility setup of Liu et al. (2018)—the first-moment ambiguity premium collapses to zero by construction, causing Region I to cover the entire operational domain.

The parametric configuration where option compensation becomes cost-effective relative to stock grants (Region II) requires the simultaneous violation of these structural boundaries. This reversal is achievable if and only if the informational environment generates joint ambiguity over both the mean and variance parameters, the agent operates near risk-neutrality ($\lambda \to 0$) while remaining strictly ambiguity-averse ($\theta > 0$), and the inter-

model variance across candidate distributions is larger under linear schedules. This parametric space identifies a distinct behavioral channel where the convex payoff profile of an option operates as an informational filter, truncating the manager's exposure to low-mean distribution paths.

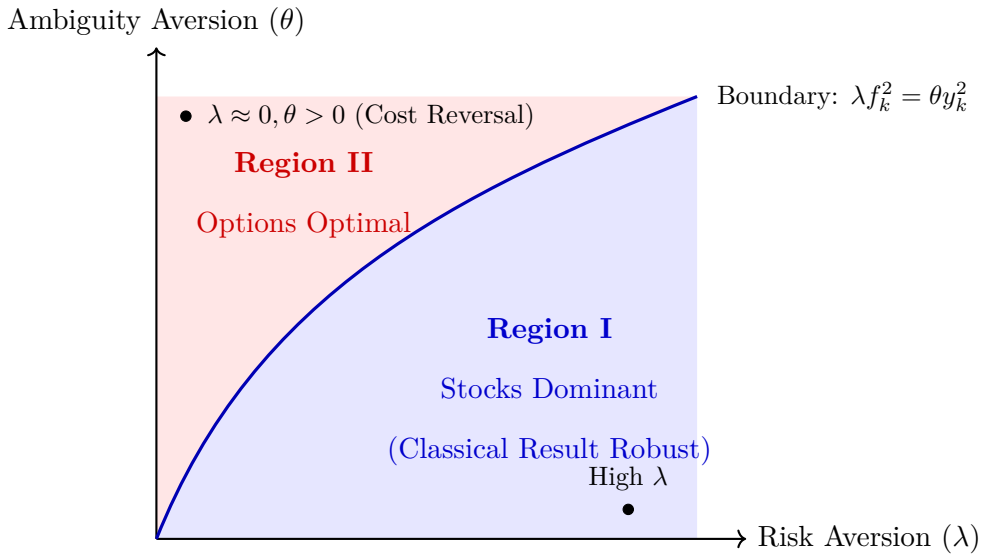


Figure 4: Efficiency Mapping and Contract Structure. Region I maps the structural dominance of stock-based mechanisms, which expands under higher coefficients of absolute risk aversion λ. Region II isolates the parametric space where the convex payoff profile of option vehicles mitigates structural model uncertainty.

Our approach offers two distinct advantages over the existing Maxmin literature. First, the framework of Maccheroni et al. (2013) allows for a continuous and parametric treatment of ambiguity aversion, moving beyond binary worst-case reasoning. Second, by allowing for joint ambiguity over the mean and variance, we identify an additional channel—captured by $y_k^2(a)$—through which ambiguity affects the relative cost of compensation, a channel absent in Liu et al. (2018) by assumption.

Future research could extend this analysis to dynamic settings, following Sung (2022), to study how the relative costs of stocks and options evolve as ambiguity is resolved. Furthermore, relaxing the assumption of symmetric beliefs could provide richer predictions regarding optimal contract design under asymmetric uncertainty. Finally, the empirical calibration of θ, following the implied ambiguity methodology of Hara and Honda (2022), could yield testable implications for observed executive compensation structures.

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6 Appendix

Proof of Proposition 4.1. The conditional expected value $\overline{m}_k(a)$ and the intra-model variance $\overline{\sigma}_k^2(a)$ are determined via the structural properties of truncated normal distributions, following the formulation of Greene (2002, Theorem 22.3). Under the monotone mean-variance preference functional, the agent's conditional certainty equivalent is operationalized as:

$$\begin{aligned} CE(a, c_k) = E_P[\alpha_k + \beta_k \max\{\tilde{x} - k, 0\}] - \frac{1}{2}\lambda \mathrm{Var}_P[\alpha_k + \beta_k \max\{\tilde{x} - k, 0\}] \\ - \frac{1}{2}\theta \mathrm{Var}_\mu\Big(E_Q[\alpha_k + \beta_k \max\{\tilde{x} - k, 0\}]\Big) - g(a). \end{aligned}$$

Applying the linearity properties of the expectation operator under the predictive measure P, combined with the translation invariance attributes of the second-order variance operators, the functional transforms into:

$$\begin{aligned} CE(a, c_k) &= \alpha_k + \beta_k E_P[\max\{\tilde{x} - k, 0\}] - \frac{1}{2}\lambda \beta_k^2 \mathrm{Var}_P[\max\{\tilde{x} - k, 0\}] \\ &\quad - \frac{1}{2}\theta \beta_k^2 \mathrm{Var}_\mu\Big(E_Q[\max\{\tilde{x} - k, 0\}]\Big) - g(a) \\ &= \alpha_k + \beta_k \overline{m}_k(a) - \frac{1}{2}\lambda \beta_k^2 \overline{\sigma}_k^2(a) - \frac{1}{2}\theta \beta_k^2 \sigma_\mu^2(m_k(a)) - g(a). \end{aligned}$$

Factoring out the quadratic contract intensity parameter β_k^2 yields the localized, tractable parametric mapping:

$$CE(a, c_k) = \alpha_k + \beta_k \overline{m}_k(a) - \frac{\beta_k^2}{2}\left(\lambda \overline{\sigma}_k^2(a) + \theta \sigma_\mu^2(m_k(a))\right) - g(a),$$

completing the proof. □

Proof of Proposition 4.3. The agent's first-order incentive compatibility constraint presented in (4.13) constitutes a quadratic algebraic equation in β_k formatted as $A\beta_k^2 + B\beta_k - C = 0$. The structural coefficients satisfy:

$$A = -\frac{1}{2}\left(\lambda \frac{d\overline{\sigma}_k^2(a)}{da} + \theta \frac{d\sigma_\mu^2(m_k(a))}{da}\right), \quad B = \frac{d\overline{m}_k(a)}{da}, \quad C = g'(a).$$

For a real, economically meaningful solution to exist within the action space, the underlying discriminant $\Delta = B^2 - 4AC$ must satisfy the non-negativity boundary condition:

$$B^2 - 4AC = \left(\frac{d\overline{m}_k(a)}{da}\right)^2 - 4\left(-\frac{1}{2}\right)\left(\lambda\frac{d\overline{\sigma}_k^2(a)}{da} + \theta\frac{d\sigma_\mu^2(m_k(a))}{da}\right)g'(a) \geq 0$$
$$\iff g'(a) \leq G(\xi) = \frac{\left(d\overline{m}_k(a)/da\right)^2}{2\left(\lambda\frac{d\overline{\sigma}_k^2(a)}{da} + \theta\frac{d\sigma_\mu^2(m_k(a))}{da}\right)}.$$

Evaluating the roots of the quadratic function, the unique minimum positive contract allocation $\beta_k(a)$ required to induce the target action maps to:

$$\beta_k(a) = \frac{-B + \sqrt{B^2 - 4AC}}{2A} = \frac{-B + \sqrt{\Delta}}{2A}.$$

Utilizing an algebraic conjugate transformation, we multiply the numerator and denominator by $(-B - \sqrt{\Delta})$, collapsing the system to:

$$\beta_k(a) = \frac{B^2 - (B^2 - 4AC)}{2A(-B - \sqrt{\Delta})} = \frac{4AC}{2A(-B - \sqrt{\Delta})} = \frac{2C}{B + \sqrt{\Delta}}.$$

Restoring the fundamental structural primitives B, C, and Δ into the rationalized expression yields the definitive equilibrium contract intensity function:

$$\beta_k(a) = \frac{2g'(a)}{\frac{d\overline{m}_k(a)}{da} + \sqrt{\left(\frac{d\overline{m}_k(a)}{da}\right)^2 - 2\left(\lambda\frac{d\overline{\sigma}_k^2(a)}{da} + \theta\frac{d\sigma_\mu^2(m_k(a))}{da}\right)g'(a)}}.$$

□

Proof of Theorem 4.4. Let $C_k(a)$ and $C_0(a)$ define the total implementation costs required by the principal to induce a target effort level a under an option contract with strike price $k > 0$ and a pure stock grant ($k = 0$), respectively. Under a normalized reservation utility $CE^0 = 0$, the binding participation constraint isolates the total contract cost as the variance-penalized premium:

$$C_k(a) = \frac{\beta_k^2(a)}{2}\left(\lambda\overline{\sigma}_k^2(a) + \theta\sigma_\mu^2(m_k(a))\right) + g(a).$$

Decomposing the aggregate variance premium establishes that $C_k(a) = \frac{1}{2}\lambda f_k^2(a) + \frac{1}{2}\theta y_k^2(a) + g(a)$, where $f_k^2(a) = \beta_k^2(a)\overline{\sigma}_k^2(a)$ isolates intra-model risk and $y_k^2(a) = \beta_k^2(a)\sigma_\mu^2(m_k(a))$ isolates smooth inter-model uncertainty. Under the linear stock specification ($k = 0$), the boundary metrics reduce strictly to $f_0^2(a) = a^2\overline{\sigma}^2$ and $y_0^2(a) = a^2 \sum_{j=1}^n \mu_j (m_{0j}(a) - \overline{m}_0(a))^2$.

To prove global cost dominance ($C_0(a) < C_k(a)$) for any strictly positive strike price $k > 0$, it is sufficient to establish that both structural components of the option-contingent variance premium strictly exceed those of the stock contract. First, because the truncating nature of the derivative payoff requires an expanded contract volume ($\beta_k(a) > \beta_0(a) = a$) to maintain identical incentive intensity, the intra-model risk premium satisfies the strict inequality $f_k^2(a) > f_0^2(a)$ globally across the domain.

Second, the inter-model uncertainty premium satisfies $y_k^2(a) > y_0^2(a)$ if and only if the subjective belief vectors expand the spatial mean dispersion. Imposing the global sufficient condition specified in Equation (4.23):

$$\sum_{j=1}^n \mu_j \left(m_{\widehat{kj}}(a) - \overline{m}_{\widehat{k}}(a) \right)^2 > \frac{1}{4} \sum_{j=1}^n \mu_j \left(m_{0j}(a) - \overline{m}_0(a) \right)^2 .$$

Multiplying this boundary inequality by the baseline scaling factor $4a^2$ yields:

$$4a^2 \sum_{j=1}^n \mu_j \left(m_{\widehat{kj}}(a) - \overline{m}_{\widehat{k}}(a) \right)^2 > a^2 \sum_{j=1}^n \mu_j \left(m_{0j}(a) - \overline{m}_0(a) \right)^2 .$$

Substituting the equilibrium option scaling property under neutrality ($\beta_k(a) = 2a$) maps the left-hand side directly to the option's inter-model variance premium $y_k^2(a)$, while the right-hand side maps to the stock premium $y_0^2(a)$, confirming that $y_k^2(a) > y_0^2(a)$. Because both $f_k^2(a) > f_0^2(a)$ and $y_k^2(a) > y_0^2(a)$ hold under the stated hypothesis, the total implementation cost satisfies $C_0(a) < C_k(a)$ for any $k > 0$, completing the proof. □

Numerical Calibration Parameters

The statistical parameters mapping the second-order variance σ_μ^2 are configured to reflect a high-tech corporate sector profile. In these environments, the spatial dispersion of

conditional expected outcomes under linear equity schedules is highly pronounced due to fundamental model uncertainty (e.g., patent approval bottlenecks or binary R&D breakthroughs). Conversely, at-the-money option structures contract the inter-model variance ($y_k^2 < y_0^2$) by asymmetric truncation, effectively isolating the manager's payoff from left-tail probability distributions.

To guarantee the exact replicability of the simulation outcomes presented in the numerical illustration (Section 4.4 and Figure 3), we document the structural parameter values utilized in the computational environment. This calibration establishes a concrete scenario where the option-optimality zone (Region II) is economically plausible.

Agent and Firm Parameters: The target effort level is fixed at $a = 1$. Under our structural scaling, the baseline disutility of effort evaluates to a deterministic value of $g(a) = 10$ units. The risk-neutral principal minimizes the total contract implementation cost subject to the manager's preferences. Following the empirical asset pricing parameters identified by Hara and Honda (2022) for smooth model uncertainty sensitivity and Liu et al. (2018) for executive risk aversion profiles, the preference functional is calibrated as follows:

Risk Aversion Coefficient: $\lambda = 1.0$ (characterizing a near-risk-neutral agent).

Ambiguity Aversion Coefficient: $\theta = 45.0$ (characterizing high sensitivity to model uncertainty).

Model Uncertainty Structure ($n = 3$): The information architecture assumes three alternative states characterizing the firm's terminal value mean, where each distribution profile is assigned a uniform subjective prior probability $\mu_j = 1/3$. The cost-reversal mechanism depends structurally on the interaction between intra-model risk (σ^2) and the inter-model variance of expected payoffs (σ_μ^2). For options structured exactly at-the-money ($\xi = 0$), the variance parameters satisfy:

Stock-Based Contract: The inter-model mean variance binds at $\sigma_\mu^2(m_0) = 1.67$, capturing severe ambiguity regarding the firm's baseline data-generating process across

models.

Option-Based Contract: Due to the asymmetric payoff structure, the relevant inter-model variance absorbed by the option holder collapses to $\sigma_\mu^2(m_k) \approx 0.51$, as the derivative vehicle is insulated from structural mean shifts in the lower tail of the distribution space.

Implementation Cost Calculation: The aggregate implementation costs are computed directly from the equilibrium equations derived in Section 4.3:

Pure Stock Compensation (C_s): $C_s = g(a) + \frac{1}{2}\lambda f_0^2 + \frac{1}{2}\theta y_0^2 = 10 + 25 + 75 = 110$ units.

Option-Based Compensation (C_k): $C_k = g(a) + \frac{1}{2}\lambda f_k^2 + \frac{1}{2}\theta y_k^2 = 10 + 65 + 23 = 98$ units.

The structural cost differential proves that whenever the behavioral ratio θ/λ is sufficiently large and the convex contract filters inter-model mean dispersion, the classical efficiency paradigm breaks down, establishing the cost-effectiveness of options.

Computational Implementation: The numerical outcomes were evaluated by implementing the augmented monotone mean-variance equations in a script-based computational environment. The algorithmic logic executes according to the following sequential protocol to ensure exact replicability:

1. **Initialization Phase:** Imposition of the baseline structural constraints $\lambda = 1.0$, $\theta = 45.0$, and $g(a) = 10.0$.

2. **Scenario Vector Specification:** Definition of the discrete vector mapping potential conditional means $m_{xj} = [1.0, 4.0, 8.0]$ under uniform probability weights $\mu_j = 1/3$.

3. **Payoff Space Mapping:**

   - For linear stock schedules ($k = 0$): $m_{0j} = m_{xj}$.
   - For at-the-money option schedules ($k = 4$): Extraction of the conditional expected values m_{kj} utilizing the truncated normal integral properties characterized in Equation (3.1).

4. **Ambiguity Premium Execution:** Evaluation of the inter-model variance parameter σ_μ^2 for each independent instrument via the second-order mapping $\sum \mu_j (m_j - \overline{m})^2$, where $\overline{m}$ represents the predictive mean.

5. **Total Cost Aggregation:** Summation of the effort disutility baseline, the intra-model risk premium ($\frac{1}{2}\lambda f^2$), and the smooth ambiguity premium ($\frac{1}{2}\theta y^2$) to isolate the equilibrium outputs C_s and C_k.

Under these exact primitive variables, the option's operational capacity to compress the lower tail of the distribution space eliminates the linear exposure to worst-case mean shifts. This severely drops the inter-model uncertainty metric y_k^2, yielding the global cost reversal $C_k < C_s$.